

Fall 2023 Project Report



Identifying travel needs, barriers, and solutions

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1. INTRODUCTION

Auto-oriented transport policies in US led to transport inequity by prioritizing the needs and preferences of car users over the broader interests of diverse communities. Consequently, current transportation system exhibits disparities in the distribution of benefits, such as access to jobs and healthcare, as well as burdens, including exposure to emissions for different population groups (Karner et al., 2016). New mobility technologies, such as automated vehicles (AVs), micro-mobility solutions, have the potential to both benefit and burden various segments of the population, creating disparities in transportation accessibility. On the one hand, smart mobility technologies like AVs could offer improved accessibility for individuals with mobility challenges, the elderly, or those who face barriers in using conventional transportation modes (Faber & van Lierop, 2020). Similarly, micro-mobility options like e-scooters or bike-sharing can enhance first- and last-mile connectivity, especially in areas underserved by traditional transit (Huang et al., 2024; Smith & Schwieterman, 2018). On the other hand, there are concerns about the equity implications of these technologies (Jiao & Wang, 2021). Affordability remains a significant barrier, particularly for electric vehicles, which might initially be more accessible to wealthier individuals. Such policies perpetuate a transportation system that disproportionately benefits individuals with access to private vehicles, inadvertently placing a disproportionate burden on various user groups. A shift toward more equitable and just transport planning is essential to address these disparities and create transportation systems that prioritize accessibility, environmental sustainability, and the diverse needs of communities.

Access to affordable, reliable, safe, and convenient transportation resources is a fundamental determinant of well-being and social inclusion. Historically and currently disadvantaged population groups often face unique travel needs and barriers that can significantly impact their daily lives. Transportation planning and infrastructure decisions, often driven by conventional approaches, tend to overlook the unique needs and challenges faced by different population groups. The dynamic nature of transportation needs, influenced by demographic shifts, technological advancements, and lifestyle changes, adds layer of complexity in measuring transport needs and barriers for transport disadvantaged. This research proposal aims to investigate and address these challenges within these communities, ultimately contributing to improved access to essential services and enhanced quality of life. Specifically, we will use mixed methods to develop and test a novel survey instrument capable of quantifying travel needs and barriers across a population. We will also provide insights into the needs and barriers faced by specific groups. The work will also involve producing policy-relevant insights that speak to the extent to which currently proposed transportation “solutions” including connected and automated vehicles, smart cities, and micro-transit can be designed and deployed to address transportation equity goals.

Existing research underscores the importance of equitable transportation access. Historically disadvantaged groups often confront challenges related to limited public transportation options, inadequate infrastructure, financial constraints, displacement, and safety concerns. Although various studies have explored these issues, there remains a need for focused research to address the unique needs of different population segments.

The study has four research objectives:

1. Identify and analyze the specific travel needs of groups that have historically and continue to currently derive disproportionately few benefits from transportation investments. These groups may include but are not limited to low-income people, disabled people, older adults, single-parent households, and Black, Indigenous, and other people of color (BIPOC).
2. Assess the physical and socio-economic barriers that hinder access to transportation services and opportunities for these population groups.
3. Design a survey instrument and analyze the resultant data to understand the factors that shape travel needs and barriers for different groups.
4. Propose practical solutions and policy recommendations to mitigate identified travel barriers and enhance access. Compare and contrast with contemporary transportation solutions.

To accomplish our objectives, we will employ a mixed-methods approach, combining quantitative and qualitative research methods. We will conduct in-depth interviews and focus group discussions to gain deeper insights into the lived experiences of the identified population groups. Following these discussions, we will design and administer a survey on travel experiences, needs, and perceived barriers. During our data collection efforts, we will work closely with community organizations, advocacy groups, and relevant agencies to identify and engage participants from historically and currently disadvantaged communities. Our goal is to ensure a representative sample and to capture a diverse range of perspectives.

2. CONCEPTS OVERVIEW

2.1 Mobility vs Accessibility

The term accessibility is frequently used alongside mobility without clear definitions or distinctions. Mobility, functioning as an intermediary service consumed for the purpose of reaching destinations, is intricately connected to the concept of travel as a derived demand (Levine et al., 2019). Travel is not an end in itself but emerges as a consequence of the underlying demand to access various destinations, highlighting the derived nature of the need for movement. According to Levine et al. (2019), improved mobility corresponds to an increase in the territory that can be covered within a given investment of time and financial resources. Accessibility, on the other hand, refers to the ease with which individuals can reach and connect with essential services, opportunities, and activities in their environment. According to Hanson (1959) accessibility measures people's potential of opportunities for interaction, contrasting with the tangible and observable aspects of revealed travel behavior. In this context, accessibility is not solely contingent on the physical movement facilitated by mobility; rather, it encompasses a broader spectrum of factors such as how closely destinations are located (proximity) and the level of connectedness within the transportation network (connectivity) (Levine et al., 2019). An improvement in accessibility implies a higher valuation of destinations that can be accessed with a given investment of time and money (Levine et al., 2019). The primary challenge lies in the lack of clarity surrounding the term "accessibility," exacerbated by its association with the concept of "universal accessibility" related to the Americans with Disabilities Act (ADA) (Handy, 2020). While measures of mobility, such as level-of-service, often simplify to the idea of vehicle delay, accessibility involves both the ease of travel and the availability and attractiveness of destinations, making it inherently more intricate (Handy, 2020; Levine et al., 2019).

2.2 Travel Need

In the realm of transportation planning literature, "travel need" is conceptualized as the underlying demand for mobility arising from individuals' and communities' requirements to access activities, services, and opportunities within their spatial environment. According to Allardt (1993), individuals are multifaceted actors engaged in diverse activity arenas to function in society and meet their needs, encompassing the dimensions of 'having,' 'loving,' and 'being.' Tonn (1984) introduces the notion that individuals prioritize activity participation and time allocation based on their beliefs about the instrumental value of activities in satisfying specific needs at a given moment. The extent to which out-of-home activities fulfill needs is contingent upon individual preferences, beliefs, and desires, which vary across persons and situations (Tonn, 1984). People's opportunities differently depending on their characteristics, aspirations, and choices (Vecchio & Martens, 2021). Ziegler and Schwanen (2011) posit that when there are changes in individuals' capacities and resources, individuals adapt their aspirations of mobility and activity needs. The degree to which individuals' needs are satisfied through out-of-home activities is determined by the dynamic interplay between individual resources (i.e., car availability) and abilities (i.e., health status), along with contextual conditions for mobility (i.e., residential location, the quality of public transport) (Nordbakke & Schwanen, 2015). This intricate interaction underscores the comprehensive influence of both personal attributes and external environmental factors on the fulfillment of individuals' needs from travel.

2.3 Travel Problem

The concept of a "travel problem" lacks an official definition in transport planning research. In our interpretation, we define travel problem as the obstacles or constraints individuals face while accessing activities, services, and opportunities due to an inadequate transport system. Travel problem stem from barriers or constraints in both the accessibility to opportunities (i.e., urban facility accessibility) and the accessibility to transport itself (i.e., public transit accessibility). Mobilizing Justice (2021) in their workshop used terms "barriers" as factors that impede the movement or access of individuals while "constraints" elements that curtail the extent to which individuals can move or access services. A range of factors might contribute travel problem such as transport system specific factors (i.e., temporal, and spatial coverage, safety, affordability, service quality), land use factors (i.e., destination types), individual factors (i.e., socio-demographic factors, reliance, skills/ability). Addressing travel problems may involve resolving specific barriers to enhance overall mobility and accessibility for individuals.

2.4 Transport Disadvantage

Transportation disadvantage afflicts people and groups that lack access to affordable, reliable, dignified, and convenient transportation options that results in lack of access to goods, services, and opportunities. Schwanen et al. (2015) in their definition highlighted several dimension of transport disadvantage which includes- a lack of access to basic services and opportunities for interaction, lack of ability or

“a relational and dynamic outcome of a lack of access to basic resources, activities and opportunities for interaction, of a lack of cognitive knowledge, know-how, aspirations and/or autonomy regarding travel and its externalities, and of a lack of influence on decision-making in the context of transport policy and governance”.

Currie and Delbosc (2011) provided a synthesis of transport disadvantaged groups identified in the literature, including those with no or limited car access, low-income individuals, women, the elderly, single parents, minority ethnic groups, youth, disabled individuals, the unemployed, beneficiaries, outer-urban dwellers, shift workers, parents traveling with children, and students. There might be situations where individuals or households willingly accept transportation challenges or limitations in exchange for other perceived benefits associated with their choice, hence transport disadvantage by choice (Karner et al., 2016;).

2.5 Transport Poverty

Mobility poverty denotes a systemic lack of transportation options, while accessibility poverty delves further by examining an individual's ability to access essential daily activities within reasonable time, convenience, and cost (Lucas et al., 2016; Preston & Rajé, 2007). Simply put, accessibility poverty refers to a state where an individual's capability to reach essential activities like work, school, healthcare, and leisure becomes so low that fulfilling basic needs becomes unattainable (Martens & Bastiaanssen, 2019; Karner et al., 2023). Lucas et al. (2016) intricately interconnect four concepts - the absence of suitable transport options (mobility poverty), challenges in reaching essential activities (accessibility poverty), financial strain due to transportation costs (transport affordability), and disproportionate exposure to adverse effects of the transport system (transport externalities) - to comprehensively define transport poverty.

2.6 Transport Related Social Exclusion

Transport-related social exclusion refers to the condition where individuals or communities experience marginalization and limited participation in societal activities due to inadequate access to transportation options (Lucas, 2012; Preston & Rajé, 2007). TRSE occurs when accessibility disadvantage reaches to a critical level (accessibility poverty) that problem of accessing opportunities prevent participation in society (Luz & Portugal, 2022). Persons facing accessibility poverty are at risk of transport related social exclusion. However, lack of accessibility is just one aspect of transport related causes of social exclusion (Kamruzzaman et al., 2016).

2.7 Transport Justice

Transportation justice, as outlined by Karner et al. (2020), embodies a normative state where individuals or groups do not face disadvantages due to a lack of access to essential opportunities required for a meaningful and dignified life. From a fairness perspective, transport policies should first and foremost address the accessibility needs of people who fall below the accessibility poverty line (Martens, 2013).

3. SYNTHESIS OF CURRENT RESEARCH ON TRAVEL PROBLEM ASSESSMENT FROM TRANSPORT DISADVANTAGED

3.1 Studies on single disadvantage people/group

People with disabilities (PWD) constitute a highly vulnerable transport disadvantage group facing frequent challenges within the transport system to meet their mobility needs. Numerous studies have explored the barriers experienced by PWD in various modes of transportation throughout different stages of their journeys. In a recent investigation by Lee et al.

(2022), Journey Mapping techniques were employed to identify travel barriers and specific needs among twenty individuals with diverse disabilities, including physical, cognitive, and visual impairments. The semi-structured one-to-one interviews revealed mode-specific challenges (related to cars, public transit, and paratransit) across different trip stages, encompassing trip planning, entering and exiting the transport. Commonly identified barriers included the need for third-party assistance, challenges with vehicle ingress/egress, difficulties in using transportation apps, lack of knowledge regarding services and schedules, and issues related to the accessibility of service locations, including wheelchair securement (Lee et al., 2022).

Prior studies have illustrated the mobility-related challenges faced by older adults as a transport-disadvantaged group, including engaging in fewer trips, covering shorter travel times, and traveling shorter distances (Wasfi et al., 2012; Collia et al., 2003). While there is extensive research on the realized mobility needs of older adults, a distinct research strand explores their unmet travel needs within the context of transport disadvantage and aging (Luiu et al., 2018; Musselwhite & Haddad, 2010; Hjorthol, 2013). Delving into the realm of unmet travel needs among older adults, the intersection of Transport Disadvantage (TD) and social exclusion has been investigated (Shergold & Parkhurst, 2012). Fields et al. (2019), in a survey involving 20 low-income older adults from two Texas cities, highlighted how transport disadvantage can result in social exclusion due to unmet and unrealized travel demands. Utilizing a custom-designed daily transportation diary app, researchers uncovered instances where the lack of suitable transport options led to missed opportunities, reduced community participation, diminished emotional well-being, lower safety, and increased burdens in terms of time and cost for the participants.

Transportation emerges as a critical need for women survivors of intimate partner violence (IPV), with Nahar and Cronley's (2021) focus group study delving into the daily mobility challenges faced by 15 immigrant IPV survivors in North Texas. The study revealed that issues such as controlling survivors' access to transportation to exert power and control, lack of independence due to not owning a car or relying on a partner for transportation, challenges in accessing grocery stores, missed healthcare appointments, feeling unsafe on public transit, and transit inaccessibility due to shelter locations being far from coverage areas all contribute to the transport disadvantage faced by IPV survivors. The possibility of encountering partners or relatives during transit journeys, longer travel times, safety concerns, and issues of reliability further compound the challenges. The authors recommend potential mobility solutions, including ride-share services (Uber, Lyft), pooled rider sharing services, and autonomous vehicles (AVs), especially for those who don't drive. Careful planning during shelter location selection for IPV survivors is also emphasized. In alignment with these findings, Leat et al. (2022), investigated the feasibility of shared mobility services in alleviating the transport disadvantage experienced by IPV survivors and their results showed that rideshare and car-share services hold promise in assisting participants in gaining social and economic independence.

Blumenberg et al. (2020) elucidated the impact of vehicle ownership status—categorized as zero-car households, car-deficit households, and fully equipped households—on various travel outcomes, including person miles traveled (PMT), vehicle miles traveled (VMT), number of trips defining activity participation, and the use of transit as an alternative mode of transport. The study discerned variations in travel patterns among populations residing in diverse residential locations with distinct racial, income, and household compositions. The findings indicate that carless households (with zero cars) and car-deficit households (with less than 1 car per driver) exhibit significantly reduced travel compared to fully equipped households (with more than 1 car

per driver). While diminished travel does not inherently imply transport disadvantage for a household, the study suggests a potential connection between not owning a car, household composition (e.g., households with children, households with disabled members), and residing in specific neighborhood types (e.g., suburban or rural areas with limited alternative transport options), which may intensify the daily travel challenges for household members. The research advocates for future investigations into how car deficiency might influence household residential location choices, employment outcomes, and activity participation.

LGBTQ+ individuals constitute transport disadvantaged group that have garnered attention in the literature due to their unique mobility needs and barriers (Zajac & Godshall, 2020; Weintrob et al., 2021). LGBTQ+ individuals may encounter barriers related to societal prejudices and discrimination, affecting their willingness to use certain modes i.e. public transport (Lubitow et al., 2017; Weintrob et al., 2021).

3.2 Studies on multiple Transport Disadvantaged people/groups

Research project conducted by Fan et al. (2023) from University of Minnesota focused on underserved or disadvantaged groups and communities in Minnesota including tribal members, Latinx, African americans, people with disabilities, immigrants, people living with HIV, single mothers, and single fathers. Utilizing a mixed-method approach, it combines qualitative interviews and quantitative smartphone surveys to comprehensively address systemic barriers and propose unique recommendations for different disadvantage groups.

Similarly, Shaheen et al. (2022) focused on unraveling the travel needs of marginalized communities dependent on Transportation Network Companies (TNCs) in the San Francisco Bay Area. By addressing the intersection of social equity and congestion relief, the report delves into the complexities of TNC usage among underserved populations, providing valuable insights that contribute to the broader understanding of transportation challenges in urban areas. Key findings include the primary motivation for pooled services being cost savings, with participants weighing these against added time considerations. The study also highlights the consideration of additional transportation modes, particularly public transit, and explores the transportation needs of people with disabilities.

Golub et al. (2019) examined equity implications of emerging "smart mobility" systems in lower-income areas of Portland, Oregon. The study, utilizing both quantitative and qualitative research methods, finds that while smart mobility systems have the potential to address transportation challenges in disadvantaged communities by lowering costs and improving services, significant barriers exist. Lower-income and racially diverse individuals face obstacles such as limited access to the smart mobility ecosystem, including bank accounts and credit cards, reliance on cash for transit, and reduced internet access. Information security concerns and disparities in internet and data usage further hinder the equitable adoption of smart mobility. Policy recommendations to address these barriers include expanding free WiFi, improving transit information, enhancing training, and providing language translation for mobile applications.

Southeastern Transportation Research, Innovation, Development and Education Center (STRIDE), as regional University Transportation Center came up with multiple studies on their recent project technical report, where researchers explored the travel characteristics and utilization of emerging mobility systems by transportation disadvantaged populations (Bardaka et al., 2022). This research reveals that while vehicle ownership is associated with higher trip rates, financial stress and spatial context significantly influence travel patterns, challenging the assumption that car ownership alone ensures equitable travel opportunities. Using NHTS data of

2001 and 2017, report findings suggest carless households and lower-income groups are more likely to use active travel, but the impact of geography and the built environment varies by socioeconomic status. The project also found how locations where transport disadvantaged population groups reside had lower rate of medical visits and recovery during pandemic, changes in transit accessibility for neighborhoods with vulnerable populations throughout the COVID-19 pandemic, investigated the utilization of Mobility as a Service (MaaS) in rural communities and assesses factors influencing current and future MaaS activity.

3.3 Travel problem measurement approach

Singer & Martens (2023) outline four distinct approaches to assess travel problems, each providing unique insights into the challenges faced by individuals in the transportation system. The first approach centers on mobility-based assessment, emphasizing factors such as travel time, cost, congestion, and reliability as key indicators. This approach delves into the level of service factors within the transport system, offering a comprehensive understanding of the mobility challenges individuals may encounter. The second approach, rooted in accessibility-based measurements, has long been recognized as a valuable tool for gauging the level of service in transportation. The third approach, focusing on travel satisfaction, has gained prominence over the last two decades. The fourth approach involves the direct measurement of travel problems, specifically targeting transport-disadvantaged groups. This approach provides a targeted examination of the challenges faced by these groups, offering valuable insights for designing interventions and policies that address their unique needs and barriers.

3.4 TRID database result

We conducted a search on TRID using the following query: “travel needs” OR “travel barriers” OR “transportation disadvantage.” Over 700 hits were returned. To narrow the search and return a more tractable number of studies, we limited the results to those published in the past five years. This approach returned 123 results when limited to “articles and papers.” We scanned the resultant titles and abstracts to identify peer-reviewed studies that explicitly focus on the travel needs and barriers faced by disadvantaged groups and that do not presuppose technological or mode-specific solutions. Our final sample of closely related work included 11 studies.

These studies generally focused on specific groups with or without an emphasis on specific modes, including transit users, people with disabilities, immigrant women of color, survivors of intimate partner violence, older adults, and car-deficit households where the number of licensed drivers is greater than the number of automobiles available. Each study is ad hoc in the sense that qualitative and/or quantitative data were collected from a generally small sample to understand transportation disadvantage conditions facing a particular group in a particular place. None of the work envisions integrating the methods into large-scale travel behavior surveys. And none simultaneously addresses how to measure and assess transportation disadvantage across a broad population.

4. UNDERSTANDING TRAVEL NEEDS AND BARRIERS FROM 2022 NHTS

4.1 Trip Outcome and socio-demographics

Age group

The life cycle generally dictates evolving travel needs, consequently influencing travel outcomes. It's evident that a teenager's activity participation rate, propensity for longer travel, and overall travel behavior differ significantly from those of older individuals. The results from the NHTS 2022 survey notably reveal that a higher percentage of older adults reported making fewer trips in the last 30 days compared to other age groups. This highlights a correlation between aging and potential challenges in travel. Beyond a simple analysis of average trip rates, lengths, and durations, the survey illuminates nuanced aspects of older adults' travel patterns. While shorter trip durations and lengths may not inherently indicate a problem, it prompts the critical question of whether this population is sacrificing access to potentially more suitable activity locations. The lower frequency of activity trips among older adults might be interpreted as a focus on local and essential activities, aligning with expectations related to retirement and potential health considerations. The key question arises: Is the lower number of trips and reduced activity participation duration problematic for individuals, regardless of their age? Addressing this requires a nuanced exploration of which specific trips are perceived as missed or forgone. Incorporating such inquiries into future surveys could yield valuable insights for crafting targeted and age-sensitive policy decisions.

Gender

Table 1 reveal that a higher percentage of women reported making fewer trips in the last 30 days compared to male. Findings underscore the nuanced differences in need and problems in travel between genders. Analyzing the average trip rate, it's notable that males reported a slightly higher frequency of trips per day (1.87) compared to females (1.79). Females, with an average trip length of 11.57 miles and a trip duration of 24.43 minutes, exhibit slightly shorter travel distances and durations compared to males, who have an average trip length of 13.53 miles and a trip duration of 27.05 minutes. Additionally, the average Dwell Time for females is reported as 96.38 minutes, while it is slightly higher for males at 110.71 minutes.

Race:

The data indicates that individuals identifying as Black/African American have the highest percentage of those reporting fewer trips (29.86%), suggesting a potential variance in travel behavior within this demographic. Interestingly, this group also reports the lowest average trips per day (1.34) compared to other racial categories. White population demonstrates a relatively lower percentage of individuals reporting fewer trips (22.67%), and they have the highest average trips per day (1.92) among all racial categories. Asians and individuals of multiple races also show unique travel patterns, with variations in trip length, duration, and Dwell Time. For instance, American Indian/Alaska Native respondents exhibit a longer average trip duration (33.34 minutes), suggesting potentially different travel needs or purposes.

Table 1: Sociodemographic of trip making outcome from NHTS 2022

Variable	Categories	Fewer Trips (%)	Average Trips/Day	Avg. Trip Length (miles)	Avg. Trip Duration (minutes)
Age	Teenager (13-18)	4.76	1.46	8.52	23.35
	Young Adult (19-44)	27.11	1.85	12.98	26.09
	Middle Adult (44-65)	28.23	2.02	14.25	26.23
	Older Adult (65+)	31.46	1.76	11.75	24.75
Gender	Male	22.02	1.87	13.53	27.05
	Female	25.44	1.79	11.57	24.43
Race	White	22.67	1.92	13.29	25.10
	Black/African American	29.86	1.34	11.31	26.36
	Asian	25.07	1.53	10.24	26.87
	AM-Ind/Al-Nat	19.00	1.44	20.15	33.34
	Nat.Haw./Pac.Isld.	17.43	1.63	7.18	42.85
	Mult. Race	16.04	1.66	7.25	23.99
	Other race	31.44	1.16	7.70	32.77
Education	Graduate	26.67	2.33	13.20	23.78
	Undergraduate	24.53	2.14	15.05	25.21
	Less than undergraduate	27.88	1.55	12.21	26.30
Work Status	Worker	26.00	2.19	13.95	25.74
	Non-Worker	31.07	1.52	11.24	25.81
Medical Condition	Yes	42.37	1.13	10.16	30.41
	No	21.70	1.93	12.82	25.28
Location	In Urban Area	23.56	1.89	11.30	24.92
	In Urban Cluster	25.58	1.79	11.12	24.24
	Not in Urban Area	23.68	1.65	17.98	29.46
Household Vehicle Count	0	34.73	1.15	5.56	37.59
	1	30.01	1.74	9.51	23.10
	2	22.37	1.92	13.46	25.50
	3	19.14	1.91	13.38	24.99
	4>=	18.48	1.80	15.54	28.77
Household Income	<=\$24,999	15.40	1.12	17.27	30.37
	\$25,000-\$49,000	19.70	1.59	10.61	25.33
	\$50,000-\$74,999	19.50	1.68	13.19	26.48
	\$75,000-\$99,999	12.90	2.08	14.04	23.94
	>=\$100,000	31.20	2.07	13.57	24.99

Source: Calculated by the authors from the 2022 NHTS

Education

Respondents with an education level below Undergraduate show the highest percentage reporting fewer trips (27.88%). While this group has a lower average trips per day (1.55), the average trip length, and duration are 12.21 miles, and 26.30 minutes, respectively.

Work Status

Workers, despite reporting more trips, may have longer travel distances and durations, possibly due to commuting or work-related travel. Non-Workers, on the other hand, report fewer trips and shorter travel durations, indicating a different travel pattern potentially influenced by varied daily activities or lifestyle choices.

Medical Condition

Respondents with a medical condition, representing individuals facing health challenges, report a higher percentage of 42.37 for fewer trips, possibly due to health-related constraints. While they have shorter travel distances and durations, indicating potentially necessary but less frequent travel for essential activities. On the other hand, individuals without medical conditions have more trips, covering longer distances, with shorter Dwell Times.

Location

Findings suggest that individuals in urban areas and urban clusters share similar travel patterns, possibly due to the proximity of amenities and services. The unique category Individuals not in urban areas exhibit longer travel distances, highlighting the impact of location on travel behavior.

Household Vehicle

Individuals from carless households report the highest percentage, with 34.73%, indicating a substantial reduction in travel activity among those without any vehicles. Comparatively, individuals from households with two or more vehicles exhibit a significant 56.52% increase in the average number of trips per day. Despite experiencing fewer trips and shorter trip lengths, individuals from carless households report a 30.28% longer average trip duration. This discrepancy could be attributed to various factors, including the nature and purpose of trips, the possibility of trip chaining, and the modes used for travel, as well as the geographic locations of activities.

Household Income

Households with incomes over \$100,000 exhibit a higher average trip rate, suggesting a more mobile lifestyle. Despite more frequent travel, shorter trip lengths and durations imply purposeful and efficient mobility, possibly influenced by better transportation options or proximity to amenities. In contrast, lower-income households travel less frequently, covering longer distances per trip. These findings highlight the intricate relationship between income, travel behavior, and lifestyle choices, emphasizing the need for tailored interventions to address diverse mobility needs across income groups.

4.2 Mode Use and Vehicle Ownership

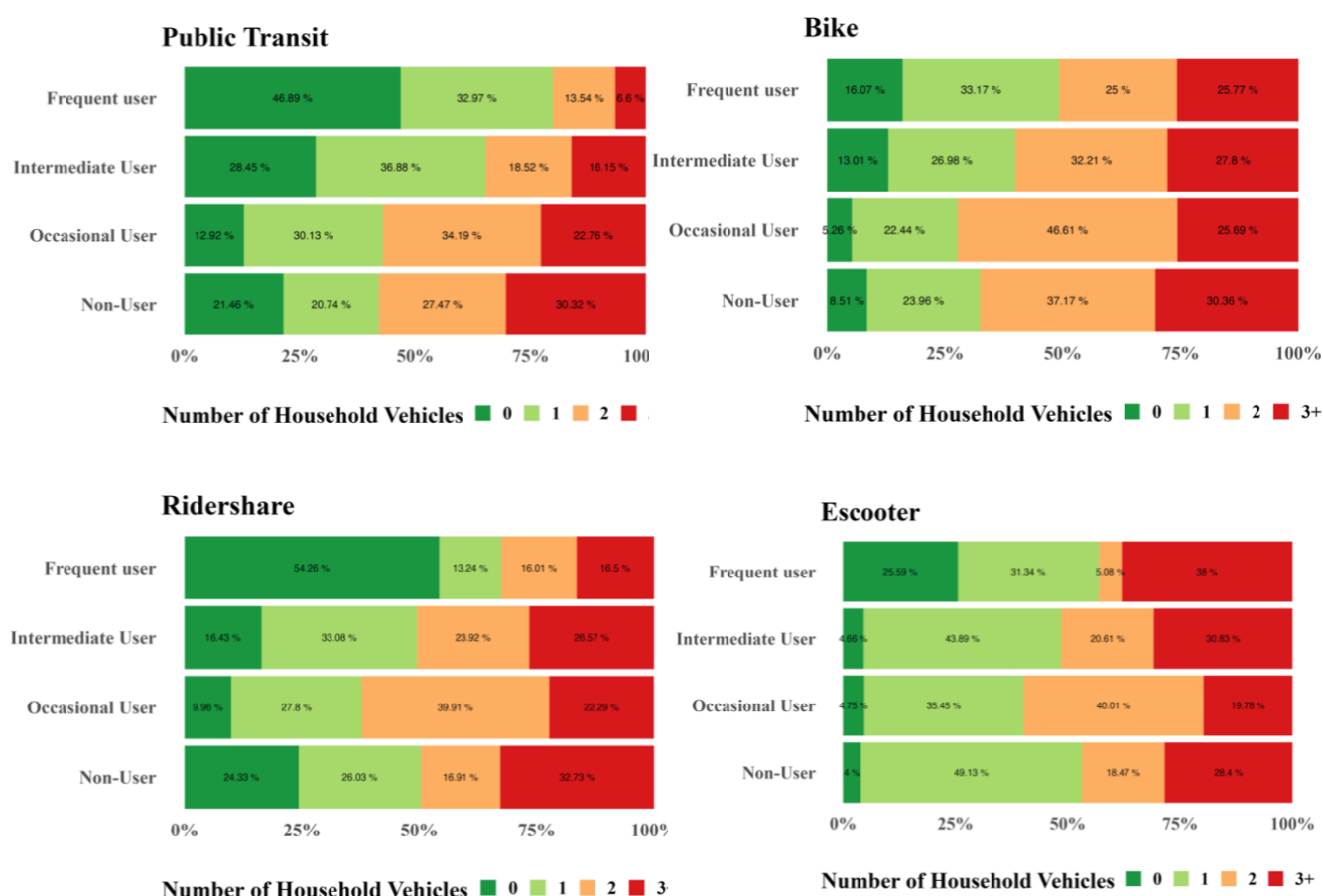


Figure 1: Household vehicle in different transport mode user groups [Source: Calculated by the authors from the 2022 NHTS]

The figure above presents a compelling insight into the relationship between different degree of mode use and their corresponding household vehicle counts. The data categorizes users into non-user, occasional user, intermediate user and frequent user based on the frequency of respective mode use in last 30 days. Individuals who are frequent users of public transit, Ridershare, Bike, and Escooter services tend to be associated with a higher proportion of households having zero vehicles. This suggests a strong correlation between active participation in various alternative transportation modes and a reduced likelihood of owning a personal vehicle within a household. The elevated share of zero-vehicle households among those frequently using Public Transit, Ridershare options, Biking, and Escooters highlights a potential shift in transportation preferences towards multi-modal choices

4.3 Trip Purpose, Day of Travel, and Household Income

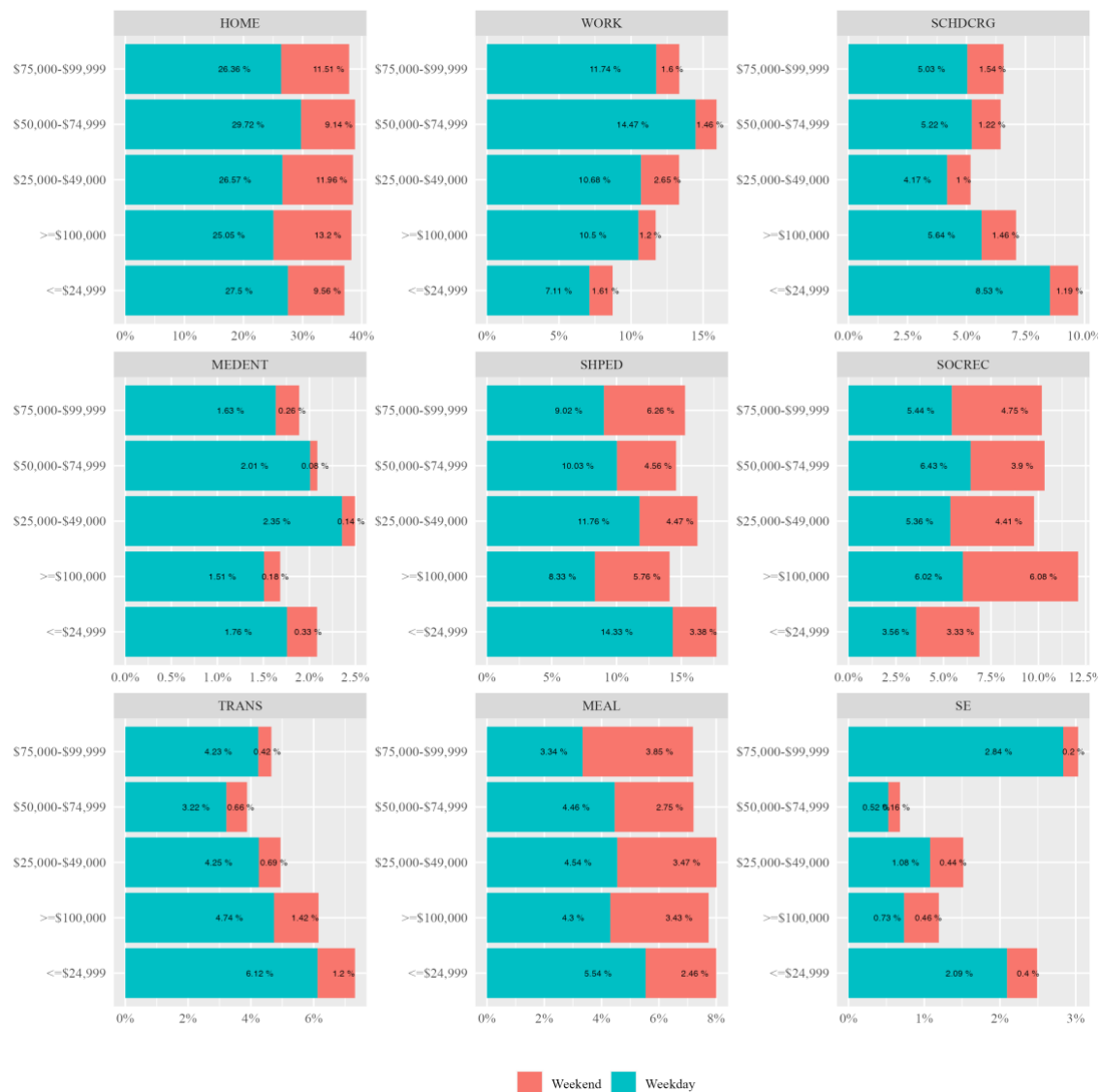
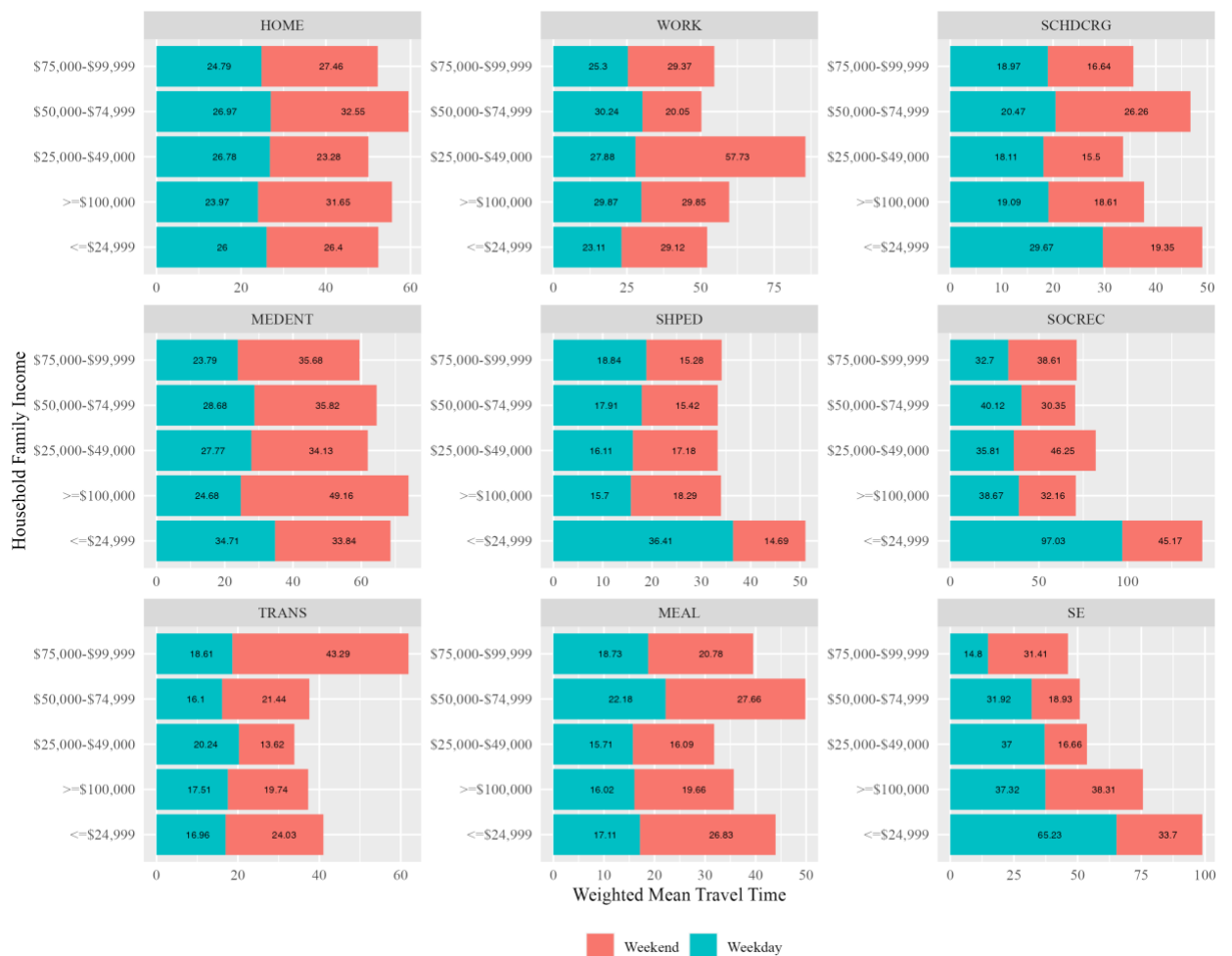


Figure 2: Percentage of different trip types by different income groups in weekdays and weekends

The data reveals a noteworthy trend where social recreation trips exhibit the lowest percentage among lower-income groups, suggesting a potential issue of transport-related social exclusion experienced by individuals in economically disadvantaged populations. The limited number of social-recreation trips could serve as an indicative marker of the challenges or financial constraints faced by those with lower incomes. This pattern is not unique to social-recreation trips; it extends to other categories such as "Transport Someone" and "School/Daycare/Religious," where individuals with lower incomes show a higher percentage compared to their higher-income counterparts.

4.4 Trip Purpose, Trip Day, and Average Travel Time



[NHTS2022 categories of Trips: 01=Home; 10=Work; 20=School/Daycare/Religious (SCHDCRG); 30=Medical/Dental services (MEDENT); 40=Shopping/Errands (SHPED); 50=Social/Recreational (SOCREC); 70=Transport someone (TRANS); 80=Meals (MEAL); 97=Something else (SE)]

Figure 3: Average travel time for different trip purposes by different income groups in weekdays and weekends

The observation that individuals in the low-income bracket tend to travel longer distances for shopping, social/recreational activities, and essential trips like school and daycare implies potential barriers in accessibility and limited local amenities.

4.5 Trip Purpose, Gender, and Average Travel Distance



Figure 4: Average travel distance by male and female of different income groups for different trip purposes

The relationship between travel distance and household vehicle ownership becomes evident in the findings, particularly noting that zero-vehicle households tend to travel shorter distances for activities like work. This trend may stem from factors such as the proximity of household work locations or a deliberate choice of residence to minimize daily commuting distances for individuals without cars. Additionally, the observation that females, on average, travel greater distances for social recreation trips compared to males (with an exception for households with four or more cars) suggests potential variations in travel behavior influenced by gender roles or preferences. Overall, the general trend indicates that, in most cases, males tend to cover greater distances on average than females, underscoring potential gender-based disparities in travel patterns.

4.6 Reasons for Fewer Trips or No trips:

Table 1 presents responses from the 2022 National Household Travel Survey of the US where the primary metric, "Fewer trips in the past 30 days", with a breakdown of reasons contributing to this reduction and corresponding response percentages. This variable is a newly added question in NHTS series which would contribute to the aspect on transport equity research. The most prominent reasons for making fewer trips include concerns related to affordability, online deliveries, health problems, reliability, safety, destination suitability and time constraints. Notably, the impact of COVID-19 is also a substantial factor. Among these, the highest percentages are observed in the categories of "health Problems", "time constraints," "more deliveries", and "unaffordability". These findings underscore the multifaceted nature of travel problems, reflecting a combination of health considerations, and economic factors influencing travel outcome. In the "other" category, encompassing unobserved factors, a substantial percentage of respondents attribute their reduced trips to unspecified reasons. This category presents an opportunity for further exploration through direct need-based assessment approach/direct measure of travel barriers.

Table 2: Reason behind fewer trips from National Household Travel Survey (2022)

Questions	YES (%)
Reason fewer trips - more deliveries	3.79
Reason fewer trips - did not feel safe	1.67
Reason fewer trips - did not feel clean/healthy	0.72
Reason fewer trips - not reliable	1.91
Reason fewer trips - did not go where needed	1.25
Reason fewer trips - unaffordable	3.85
Reason fewer trips - health problems	3.95
Reason fewer trips - no time	4.18
Reason fewer trips - other	7.7
Reason fewer trips - COVID 19	6.20

The largest proportion of respondents falls into the category of "Valid skip" with 62.32%, indicating a substantial portion of the surveyed population did not provide specific reasons for making fewer trips, possibly due to the nature of the survey or personal preferences. The options of "I prefer not to answer," "Don't know," and "Not ascertained" contribute to 5.82%, 6.22%, and 2.81%, respectively, showcasing the presence of respondents who may choose to withhold information or may genuinely be uncertain about the reasons for reducing their travel.

Table 3 shows more specific reasons for making fewer trips include "Personally sick or quarantining" (2.02%), "Vacation or personal day" (1.61%), "Worked at home (for pay)" (2.51%), and "Bad weather" (2.04%). These categories shed light on health-related, personal, and environmental factors influencing travel behavior. Additionally, options like "Household chores/projects" (1.55%) and "Caretaking" (0.55%) indicate that domestic responsibilities and caregiving duties play a role in travel decisions.

Table 3: Reasons for not taking trips.

<i>Options</i>	<i>Sample</i>	<i>Percentage</i>
-1=Valid skip	10,592	62.32
-7=I prefer not to answer	990	5.82
-8=Don't know	1,057	6.22
-9=Not ascertained	477	2.81
01=Personally sick or quarantining	343	2.02
02=Vacation or personal day	273	1.61
03=Caretaking	93	0.55
04=Disabled or home-bound	144	0.85
05=Worked at home (for pay)	427	2.51
06=Not scheduled to work	130	0.76
07=Household chores/projects	263	1.55
08=Bad weather	347	2.04
09=Out of country	4	0.02
10=No transportation available	92	0.54
12=School holiday/teacher workday	58	0.34
14=Hospitalized or otherwise confined	22	0.13
15=School from home	111	0.65
16=Reluctant to travel	83	0.49
97=Something else, please specify	1491	8.77

N= 16997

Source: Calculated by the authors from the 2022 NHTS sample data (without weighting)

Table 4 sheds light on the impact of conditions or disabilities on travel behavior among respondents. Notably, around 2.09% of participants reported restricting their driving to daytime due to a condition or disability. The table also reveals that respondents frequently explored alternatives like asking others for rides (2.59%), giving up driving (1.84%), and reducing travel (5.84%) due to conditions or disabilities. However, the use of public transportation, rideshare services, and special transportation services due to conditions or disabilities was comparatively low, each ranging from 0.45% to 0.51%. These findings collectively illuminate the varied ways in which conditions or disabilities influence travel choices, emphasizing the importance of accommodating diverse needs in designing transportation systems.

5. NEED FOR FUTURE RESEARCH

National Household Travel Surveys (NHTS) and similar national-scale surveys play a crucial role in understanding travel patterns and demand across diverse demographic groups. However, their primary focus on quantifying travel demand often falls short in capturing the nuanced travel needs and challenges experienced by various socio-demographic segments. These surveys may not comprehensively identify the spectrum of travel problems faced by individuals,

especially who are transport disadvantaged. Issues such as accessibility poverty, where certain groups lack easy access to essential services due to transportation limitations, and variations in activity participation are often overlooked. Furthermore, the impact of these constraints on the overall well-being and travel satisfaction of different demographic cohorts is not sufficiently addressed. To address these gaps, there is a growing need to augment traditional travel surveys with a qualitative exploration of individuals' experiences, incorporating a broader lens that considers not only the quantity of travel but also the quality, equity, and social implications of the journeys made. This expanded approach is vital for crafting inclusive transportation policies that cater to the diverse and often unmet needs of various transport disadvantaged groups.

Categorizing Travel Problems for future Survey:

Following travel problem can be asked in survey from general population groups to get sense of transport disadvantage conditions:

- Transport Infrastructure: signs/signals, information display, ticketing systems, directness of routes, shelter design, language barriers, poor sidewalk, poor aesthetics, no/inadequate streetlights
- Service quality: reliability, cleanings, comfort, transfers time and frequency, waiting time, staff friendliness.
- Affordability: High cost of fares, Price of vehicle (i.e., electric vehicle, scooter price)
- Safety: Evening/night travel, darkness level at street/sidewalk, fear of crime, harassment, police presence
- Temporal coverage: departure time, transit schedules/frequency, opening hours.
- Spatial coverage: distant destination locations, unmatched destination type, no public transit, first and last mile
- Modal mismatch: forced use of specific mode, trip chaining capacity, alternative mode availability.
- Reliance: compromising independence, social network
- Skill/ability: can't drive/ not a skillful driver, can't rider scooter/bike, technological literacy, optimization/planning capacity.
- Preferences/Perception: EV/AV/Micro mobility is unappealing, psychological issues, no time to travel/don't like to travel, prejudice

Categorizing direct impacts of travel problems for future Survey:

Following travel problem can be asked in survey from general population groups to get sense of accessibility poverty/social exclusion/overall quality of life:

- Forgone activities: cut leisure time, cut recreational activities, limiting search radius for jobs.
- Delayed activity participation: late at school, late at job, late for hospital appointment, late at social events
- Wellbeing: stress and frustration, helplessness, satisfaction from travel
- Financial strain: paying more on transportation,
- Access trips: more VMT/VKT,

- Benefit (although accepted travel problem): (a) affordable housing (b) high salaries (c) safety (d) childcare facility on route (e) school quality (f) work flexibility (g) healthcare (h) social environment
- Virtual access: Utilizing delivery services, online shopping, engaging in telemedicine, opting for virtual social activities,
- Long term trip making behavior adjustment: modifying travel frequency, drive fewer miles, make less trips, combine trips, costly ride-sharing service use, working from home, arranging carpools, using public transportation.

Categorizing/Identifying socially disadvantaged person/groups/place:

Following population groups can be crucial to understand transport disadvantage situation:

- Age (Youth, Older Adults)
- Gender (Female, LGBTQ+)
- Race and Ethnicity (Black/African American, Latinx, Immigrants)
- Education (Illiterate People)
- Income (Low Wage Earner)
- Employment. (Unemployed)
- Physical Condition (Disable/Poor Health, medical condition, pregnant)
- Car Ownership (No car households)
- Internet access (No internet/costly internet package use at home or phone)
- Household Type (Single Parent)
- Built environment/Land use: sprawl, rural/suburban location, low density suburb.
- Inefficient Location: High housing+ transport cost, less diversity of opportunities,

Sample Question: Focusing on unrealized but desired demand (reasons, importance, consequence) on health, emotional wellbeing, overall quality of life.

- Have you experienced any of the following travel barriers/constraints?
- How frequently do you encounter difficulties in participating in your activities due to travel barriers/constraints?
- How often do you find yourself spending more time or traveling more distance than you would prefer?

6. PROJECT PLAN

6.1 Anticipated Outcome

This research project anticipates yielding a diverse range of outputs that promise to contribute to our understanding of transportation equity and provide practical solutions for addressing access barriers. These outputs encompass:

- Publications and Conference Papers: 1-2 peer-reviewed academic publications and associated conference papers including one comprehensive literature review and one reporting on mixed-methods analysis and results.

- **Policy Paper:** One policy paper that translates our research into actionable recommendations for decision-makers and agency staff. These papers will outline strategies for enhancing transportation equity.
- **Web-Based Resources:** We will establish dedicated websites and online resources that centralize our research findings, making them accessible to a wide audience, including community organizations, transportation planners, and the general public. These platforms will serve as hubs for sharing insights and best practices.
- **Survey Instrument:** A novel and robust survey instrument will be developed and shared, enabling other researchers and organizations to quantify travel needs and barriers across different populations. This tool will facilitate future studies on transportation equity.
- **Practical Solutions and Policy Recommendations:** As a core objective, this research project will offer practical solutions and policy recommendations to mitigate identified travel barriers.

6.2 Analysis plan

Qualitative data from interviews and focus groups will be subjected to thematic analysis and inductive coding to extract valuable insights, categories, and personal narratives. Quantitative data will be analyzed using statistical modeling techniques to identify trends and patterns in travel needs and barriers. We will validate our survey instrument using standard approaches that assess face validity and internal consistency of the relevant constructs, including expert review and principal components analysis. We will seek to use as few questions as possible to identify people and households experiencing unmet needs and barriers so that our methods can be applied by others.

We anticipate that this research will shed light on specific travel needs, barriers, and experiences within historically and currently disadvantaged communities. The findings will contribute to a better understanding of transportation equity issues and will inform the development of targeted interventions and policies to address these challenges that may differ in important ways from approaches currently in vogue.

7. CONCLUSION

This research holds significant implications for both basic and applied work. By identifying and addressing travel needs and barriers within historically and currently disadvantaged groups, we aim to promote equitable access to essential services, employment opportunities, education, and healthcare. This, in turn, can foster greater social inclusion and improved well-being for these communities. By leveraging a mixed-methods research approach, we will generate comprehensive insights that can inform evidence-based policy recommendations and interventions. Ultimately, this research strives to contribute to a more equitable and inclusive transportation system that connects people to the opportunities they need to lead a meaningful life.

The desired impacts of the work include transportation policy, planning, decision-making, and investments that directly address the needs of populations that have historically been and/or are currently underserved by the system. Understanding and addressing these needs is a critically important for achieving transportation equity. We will communicate that that

transportation solutions are not one-size-fits-all, but rather tailored to the unique challenges faced by disadvantaged groups. The project could ultimately reconfigure the transportation landscape, placing historically marginalized and underserved communities at the center of all aspects of transportation planning and decision making.

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